RFID-Based Automated Parking System for Revenue Transparency at BzBlock Parking Facility

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ABSTRACT

Title: **RFID-Based Automated Parking System for Revenue Transparency at BzBlock Parking Facility**

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This study was conducted to address persistent revenue-related issues in traditional parking operations that rely on manual recording, verbal confirmation, and paper-based payment monitoring. Such practices often result in unpaid exits, inconsistent fee computation, reuse of previous payments, and limited transparency in daily parking income. These challenges significantly affect financial accountability and hinder effective management of parking facilities. To respond to these concerns, the proponents developed an

RFID-Based Automated Parking System for Revenue Transparency at the BzBlock Parking Facility.

The proposed system utilizes RFID cards to record vehicle time-in and time-out, automatically compute parking duration, and generate accurate parking fees. All transactions are digitally stored and displayed through a web-based dashboard accessible to the parking owner and managers. The system ensures proper payment validation by requiring each transaction to be marked with a verified payment status, thereby undocumented revenue.

A mixed-method research approach was employed to assess the existing parking conditions. Qualitative data were gathered through direct observation and structured interviews with the six parking personnel to identify recurring operational and revenue-related challenges. Quantitative data were collected by documenting the frequency of incorrect fee computation, and transaction inconsistencies. The combined findings confirmed that manual systems contribute significantly to revenue loss and monitoring difficulties.

The results indicate that implementing an RFID-based automated parking system can improve transaction accuracy, strengthen revenue transparency, and provide reliable financial records for management decision-making. The study demonstrates that automation focused on payment validation and record management can serve as an effective solution for enhancing accountability and sustainability in small- to medium-scale parking facilities.

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CHAPTER 1

INTRODUCTION

A. BACKGROUND OF THE STUDY

Parking management has become a persistent operational challenge in many commercial establishments, particularly in multi-tenant buildings that accommodate a high volume of daily customers. While parking facilities are intended to generate additional income and improve customer convenience, inefficient payment monitoring and weak revenue tracking often result in financial losses. These issues are commonly observed in establishments that rely on manual parking operations, where limited transaction documentation and human-dependent processes reduce revenue transparency.

One establishment currently experiencing this challenge is the BzBlock Building, a developing commercial structure officially founded on July 24, 2020, during its blessing ceremony. The building was established by Mr. Elmer V. Diaz, whose vision was to create a dynamic and activity-filled commercial hub hence the name “BzBlock,” derived from the word *busy*. Although the building is still progressing toward full operation, it was designed with long-term sustainability in mind, including the provision of an organized parking facility to support tenant and customer activities.

BzBlock is located at 1235 Tungkong Mangga, City of San Jose del Monte (CSJDM), Bulacan. It accommodates various business establishments such as BPO offices, beverage and refreshment stalls, retail shops, and massage and spa services, with its primary business function focused on real estate leasing. At present, the building employs six staff members, six of whom are assigned as parking personnel responsible for managing the two-floor parking facility with 82 total parking slots 55 for motorcycles and 27 for cars.

Despite continuous development, the parking facility experiences a significant problem related to revenue loss. Under the existing manual parking setup, payment collection and transaction recording depend entirely on parking staff. In several instances, vehicles are able to exit the premises without completing proper payment.

Another recurring issue arises when a customer parks within the allotted time they have already paid for and leaves without exceeding that duration. When the same customer returns on the following day, parking staff may mistakenly allow the vehicle to park for free, assuming that the previous payment is still valid. The absence of a centralized digital transaction system also limits the building owner’s ability to monitor daily parking revenue accurately. Parking fees collected manually are not automatically logged, making it difficult to validate income, review transaction histories, or detect discrepancies. As a result, parking revenue often becomes insufficient to fully support operational costs such as staff salaries, equipment maintenance, and system improvements. Studies indicate that manual parking systems are highly vulnerable to revenue leakage due to inconsistent fee enforcement and lack of transparent transaction records (TechDay, 2023).

Furthermore, the reliance on manual computation of parking duration increases the risk of miscalculation and unintentional undercharging. Without automated time-in and time-out recording, accurate fee assessment becomes challenging, especially during peak hours when staff manage multiple vehicles simultaneously. Research highlights that automated payment systems significantly improve revenue accuracy and accountability by ensuring that all transactions are properly recorded and verified (MatrixComSec, 2024).

To address these concerns, this study proposes the RFID-Based Automated Parking System for Revenue Transparency at BzBlock Parking Facility. The proposed system utilizes RFID technology to automate time recording, payment validation, and transaction documentation while maintaining staff involvement in card distribution and payment confirmation.

Upon entry, customers proceed to the parking staff, who scan and issue an RFID card. The scanning process automatically records the time-in of the vehicle in the system. All active parking entries are reflected in real time on the owner's and staff's dashboard, providing immediate visibility of ongoing transactions.

During exit, the RFID card is returned to the parking staff for scanning. The system automatically calculates the total parking duration and corresponding fee. Only transactions marked with a "paid" status will allow the exit process to proceed. All transactions are stored in a digital log, enabling accurate revenue tracking and audit review.

II. REVIEW OF RELATED LITERATURE, STUDIES AND SYSTEM

This chapter presents the relevant literature and studies that support and consider in strengthening the importance of the present study.

A. REVIEW OF RELATED LITERATURE

Zhang (2021) presented a method for automatic parking systems that utilizes a combined algorithm consisting of bidirectional Breadth-First Search (BFS) and a modified Bellman–Ford algorithm to determine the shortest parking path within complex environments. Although the study primarily focuses on automated vehicle navigation rather than barrier or card-based systems, it highlights the importance of automation in reducing human intervention and improving operational efficiency. The research emphasizes how automated processes can enhance accuracy and system reliability, which supports the concept of minimizing manual involvement in parking operations.

Co et al. (2022) examined the effectiveness of multi-level car parking structures as a solution to traffic congestion and parking shortages in Davao City. Their study revealed that increasing vertical parking capacity and improving parking facility design can significantly reduce roadside congestion and promote more organized parking operations. access systems, such as controlled entry and exit mechanisms, in improving overall parking management efficiency, which aligns with the objectives of the proposed automated system.

Dichoso Jr. et al. (2022) conducted a study assessing the feasibility of implementing an automated parking system using RFID technology in Tanauan City, Batangas. The developed system automated three major parking processes: vehicle entry and exit and payment processing. The results showed that automation significantly reduced human error, improved operational efficiency, and achieved a high profit ratio of 68% with a payback period of only three years. This study strongly supports the present research, as it demonstrates that RFID-based parking systems are both practical and financially viable within the Philippine context.

Myint and Oo (2022) presented a multi-storey parking system utilizing RFID technology, a PIC microcontroller, and infrared (IR) sensors. In their study, both vehicle entry and exit, as well as parking slot availability, were managed through RFID cards and sensor-based detection. The results demonstrated that the integration of RFID is a reliable and effective solution for managing vehicles in large parking facilities.

Wang et al. (2022) proposed a barrier-free access control system using ultra-high frequency (UHF) RFID technology, eliminating the need for physical flap barriers and manual card-swiping procedures. Their system employed a dual-antenna tag-array model capable of detecting motion direction and identifying illegal intrusions. Experimental results showed high accuracy rates of 99.83% for direction detection and 96.67% for intrusion identification. This study is relevant to the present research as it demonstrates the effectiveness of RFID technology in automating entry and exit processes, minimizing human interference, and improving transaction efficiency—concepts that align with the proposed RFID-based parking system with smart barrier control.

Yahya et al. (2022) explored the integration of RFID technology with cloud computing and Internet of Things (IoT) platforms to support smart parking systems. Their framework enabled parking slot reservation, real-time status updates, and controlled vehicle access using RFID authentication. The study emphasized that RFID-based monitoring can reduce system uncertainty and ensure that only authorized users are allowed to enter and exit parking facilities. These concepts are directly applicable to the present study, particularly in improving the reliability of parking transaction records.

Elfaki et al. (2023) developed and evaluated a smart real-time parking system that integrates artificial intelligence (AI), IoT technologies, license plate recognition, and automated payment mechanisms. The system aimed to optimize parking slot utilization, reduce traffic congestion caused by vehicles searching for available spaces, and improve exit processing efficiency. Although the study primarily focused on real-time monitoring and slot allocation rather than card-based entry systems, it reinforces the importance of automation in enhancing parking accuracy, reducing operational delays, and supporting efficient payment management.

Luciano et al. (2023) designed and implemented a web-based parking management system known as PARADAJuan, using multi-paradigm programming languages including HTML, CSS, JavaScript, PHP, MySQL, and Flutter. The system focused on digital recordkeeping for vehicle entry, exit, and parking fee computation in local parking facilities in the Philippines. The findings revealed that automated record management improves data accuracy, reduces manual workload, and enables parking owners to monitor income more effectively. Although the system did not incorporate RFID cards or automated barriers, it shares the same objective as the current study—minimizing human intervention while enhancing revenue tracking and management efficiency. Consequently, this research provides strong local evidence supporting the effectiveness of computer-based parking automation.

Cruz et al. (2024) utilized camera-based systems and image recognition technology to identify available parking slots and monitor vehicle movement in real time. Their findings indicated that the system enabled drivers to locate vacant spaces more efficiently and reduced carbon emissions caused by prolonged vehicle idling while searching for parking areas. Although the study employed computer vision rather than card- or RFID-based access systems, both approaches emphasize the use of technology to enhance operational accuracy and minimize dependence on manual monitoring.

Trinidad and Porquis (2024) conducted a study focused on the development of a monitoring device designed to detect illegally parked vehicles. The research addressed issues related to parking discipline, enforcement, and real-time supervision in order to maintain smoother traffic flow and promote responsible vehicle behavior. Their findings underscored the importance of automated monitoring systems in regulating parking activities, reducing violations, and improving management efficiency. This study supports the present research, which likewise aims to strengthen parking operations through automation—particularly in preventing unauthorized entry or exit.

B. REVIEW OF RELATED STUDIES

Dichoso et al. (2022) conducted a feasibility study on an RFID-based automated parking management system for a commercial site in Batangas, Philippines. The study focused on automating vehicle entry, payment, and parking operations to reduce manual processes. Results showed that the proposed system is feasible and can improve efficiency in parking management. The researchers concluded that RFID technology can minimize human intervention and reduce errors in parking operations.

Channamallu et al. (2023) conducted a comprehensive review of smart parking systems (SPSs) by analyzing 124 research papers to understand current developments and trends in parking technologies. The study found that SPSs address inefficiencies such as congestion and poor revenue management by optimizing space utilization and improving user experience. IoT-based solutions and wireless sensor networks were identified as commonly used technologies due to their cost efficiency and decision-support capabilities. The study also discussed challenges and future research directions for smart parking systems.

Tan et al. (2023) designed an IoT-based smart parking management system implemented in an urban area in Manila. The study aimed to improve parking operations by automating parking processes and reducing congestion. Findings revealed that the system helped organize parking activities and improved overall efficiency. The researchers concluded that IoT-based parking systems are useful in managing parking in busy urban areas.

Maristela et al. (2024) evaluated an IoT-based parking and monitoring system implemented at Lyceum of the Philippines University–Batangas. The study aimed to improve parking organization and reduce congestion within the campus. Results showed that the system helped manage parking spaces more efficiently. The researchers concluded that automated parking systems are beneficial for campus parking management.

Ramos et al. (2024) developed an automated parking system using UHF-RFID technology integrated with IoT devices and a cloud-based application. The system was designed to automate vehicle entry and accurately record parking transactions. Based on ISO/IEC 25010 evaluation results, the system performed well in terms of functionality, usability, and reliability. The study showed that RFID-based systems can improve parking efficiency and management transparency.

Ahmed and Chaudhari (2025) proposed a smart RFID-based automated car parking system designed to reduce manual intervention and improve parking efficiency in urban areas. They concluded that the system effectively automates vehicle entry and exit, enhances space utilization, and supports better parking management through secure and reliable RFID technology.

Ingle et al. (2025) developed an IoT-based car parking system using RFID to automate vehicle identification, access control, and parking operations. The system integrates RFID tags and readers with microcontrollers, sensors, and a cloud database to reduce manual intervention during entry and exit. The study concluded that the system improves efficiency, security, and user convenience while supporting transparent parking management. It also noted that the system can help reduce operational costs.

Muhamad et al. (2025) proposed *Cyberpark*, an IoT-based automated parking management system that integrates RFID technology with microcontrollers to automate vehicle access and improve parking operations. The system uses RFID readers for secure entry and exit along with cloud connectivity for centralized data management. The researchers concluded that the system effectively automates gate operations and enhances overall parking efficiency. The study showed that RFID- and IoT-based systems are suitable for scalable parking management.

Soria (2025) studied the implementation of an RFID-empowered parking system within a university campus. The system aimed to improve access control and parking efficiency for students and staff. Findings showed that RFID technology enhanced parking security and reduced manual checking. The study concluded that RFID-based parking systems are effective for improving parking management in educational institutions.

C. REVIEW OF RELATED SYSTEMS/PROJECTS

Figure 1 : Automated Smart RFID Card Ticket Dispenser Machine with Car Parking Management System

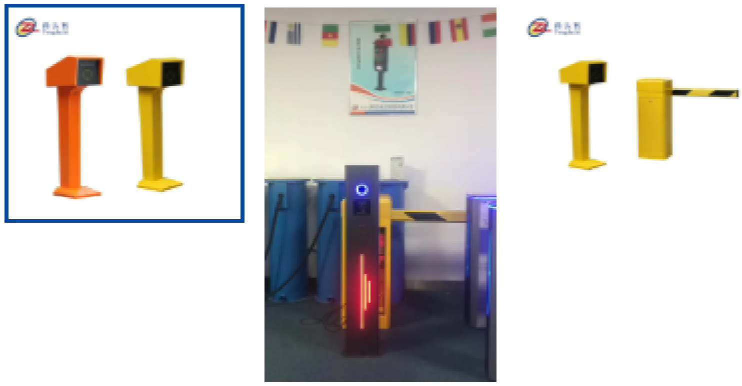


Figure 2 : TDZ - Full Automatic Rfid Smart Card Reader Car Parking System

For Vehicle Access

Figure 3 : Automatic Card Dispenser Machine With Payment Terminal Parking

System

Figure 4 : RFID Based Access Control

Figure 5: Parking System and Technology

III. STATEMENT OF THE PROBLEM

A. GENERAL PROBLEM

The absence of an automated parking system that ensures accurate payment validation and transparent revenue recording at the BzBlock Parking Facility. Some vehicles are allowed to exit without settling their parking fees. This lack of automated verification results in revenue leakage, inconsistent transaction records, and difficulty in monitoring actual parking income.

B. SPECIFIC PROBLEM

1. When the same customer returns the following day, the parking staff may mistakenly allow the vehicle to park for free, assuming that the previous day's payment is still valid because the customer did not exceed the paid parking time.
2. The owner is not aware that the parking staff continues to micromanage certain situations that are no longer included in the established parking policies.
3. The absence of transaction logs makes it difficult for the parking owner to monitor daily parking income and detect revenue discrepancies.

IV. OBJECTIVES OF THE STUDY

A. GENERAL OBJECTIVE

The general objective of this study is to design and develop an RFID-Based Automated Parking System for Revenue Transparency at the BzBlock Parking Facility that ensures accurate payment validation and provides reliable transaction records to support efficient parking revenue management.

B. SPECIFIC OBJECTIVE

1. To automatically validate parking transactions on a per-day basis using RFID technology in order to prevent the reuse of previous parking payments.
2. To implement a system-monitored parking process that enforces established parking policies and reduces unauthorized staff decision-making.
3. To generate a centralized digital transaction log that enables the parking owner to monitor daily parking income and detect revenue discrepancies.

V. HYPOTHESIS

H1: If an RFID-based automated parking system is implemented, revenue records will become more transparent, and parking operations will be more organized compared to the current manual system.

H0: If an RFID-based automated parking system is implemented, it will not significantly improve revenue transparency, or enhance parking operations.

VI. THEORETICAL FRAMEWORK

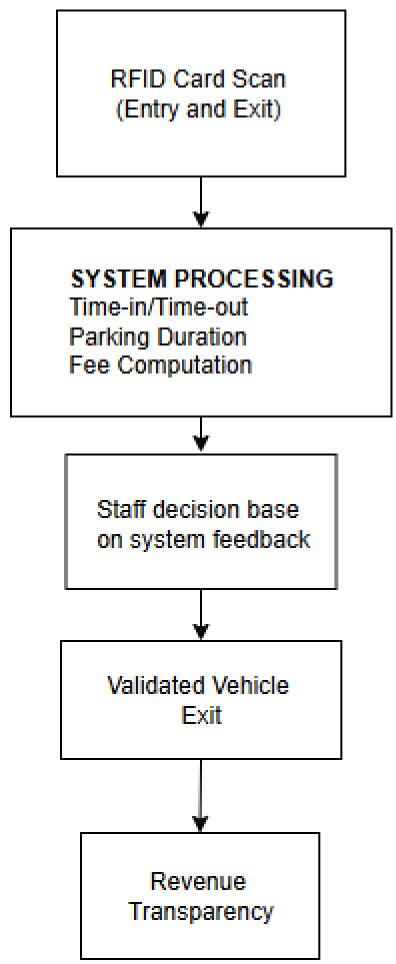


Figure 6: Control Theory

The theoretical framework of this study is anchored on Control Theory, which explains how systems maintain accuracy and reliability through continuous monitoring and feedback mechanisms. In the proposed RFID-Based Parking System for Revenue Transparency at the BzBlock Parking Facility, the process begins with RFID card scanning conducted by parking staff during vehicle entry and exit. The scanned data serves as the system input, recording the time-in and time-out of each vehicle. The system then functions as the control mechanism by computing the parking duration, calculating the corresponding fee, and verifying the payment status of each transaction. Through its feedback mechanism, the system displays whether the transaction is marked as paid or unpaid, guiding the staff's decision to allow or restrict vehicle exit. This system-based validation replaces manual judgment and prevents the reuse of previous payments, unpaid exits, and inaccurate charging.

VII. CONCEPTUAL FRAMEWORK

INPUT

KNOWLEDGE REQUIREMENTS:
- Android programming language related
- Database management
- User Interface and User Experience

SOFTWARE REQUIREMENTS:
- Javascript
- React, Express
- Postgresql
- Visual Studio Code
- Arduino IDE
- Cloud-based server

HARDWARE REQUIREMENTS:
- Arduino microcontroller (ESP32 CP2102)
- RC522 RFID Reader
- RFID CARD 13.56MHz
- LCD Display I2C

PROCESS

ITERATIVE WATERFALL MODEL

SYSTEM REQUIREMENTS
- Analyze problems
- Specify goals and scopes
- Identify potential clients
- Conduct on-site observation of current parking condition
- Interview client for insights and concerns

SYSTEM ANALYSIS & DESIGN
- Analyze system requirements
- Design flowchart
- Design data flow diagram
- Design system architecture
- Design database

SYSTEM IMPLEMENTATION
- Assemble hardware components
- Develop the software (web or mobile dashboard)
- Integrating all Components

SYSTEM MAINTENANCE
- Scanner Accuracy
- System Load and Testing
- Feedback from Trial Users

OUTPUT

RFID-Based Automated Parking System for Revenue Transparency at BzBlock Parking Facility

BZ PARKPAY ADMIN WEB APP

DOCUMENT

USER'S MANUAL

FEEDBACK

Figure 7: Conceptual Framework

The input phase consolidates everything that is required of system development and includes the knowledge and requirement for IoT-based, mobile apps (Android and iOS) and cloud computing with database management. This also includes software requirements like JavaScript, React, Express and MySQL and various tools like Visual Studio, and the Arduino IDE. The hardware requirements for this are listing of Ultrasonic sensors, and the Arduino boards to be used, cloud-based servers, and mobile devices for user interaction.

This study adopts the Iterative Waterfall development model for the design and implementation of the BZ PARKPAY System: An RFID-Based Automated Parking System for Revenue Transparency at the BZ Block Parking Facility. The system is developed through a structured and sequential process, consisting of requirements analysis, system design, implementation, testing, deployment, and maintenance. Each phase is completed and documented before proceeding to the next to ensure clarity, consistency, and correctness of system functions such as RFID-based card usage, transaction recording, and revenue reporting. The primary outcome of this methodology is improved revenue transparency and accountability at the BZ Block Parking Facility. The system’s implementation and long-term use are supported by comprehensive documentation and a user manual, enabling administrators and stakeholders to efficiently operate and manage the BZ PARKPAY system through a streamlined and easy to use interface.

VIII. SCOPE AND DELIMITATION

SCOPE

Parking Managers – Parking managers will be the primary system users responsible for daily parking operations. They will utilize the web-based dashboard to record vehicle entry and exit through RFID card scanning, verify payment status, and ensure that all transactions are properly logged. The system will assist parking managers in computing parking fees accurately, marking payment status, and preventing the reuse of previous payments. Through the automated records, managers can reduce manual errors and ensure that all parking transactions are properly documented.

Parking Owner – The parking owner will have full access to the system's revenue monitoring features. The dashboard will provide summarized and detailed transaction reports, including daily, weekly, and monthly parking income. The owner can review complete transaction logs, monitor payment consistency, detect revenue discrepancies, and evaluate overall financial performance. This access supports transparency, accountability, and informed decision-making regarding parking operations and income management.

DELIMITATION

1. The system supports parking fee computation and payment recording only through the designated parking process implemented by the facility. Other payment platforms such as mobile wallets, online banking, or third-party applications are not included in the scope of this study.
2. The system focuses on ensuring accurate transaction recording, payment validation, and revenue transparency. It does not cover customer disputes regarding parking rates, discounts, penalties, or special exemptions such as senior citizen privileges.
3. The study is limited to the monitoring and reporting of parking transactions for management and ownership purposes and does not include vehicle guidance systems, slot availability displays, or traffic flow optimization features.

IX. SIGNIFICANCE OF THE STUDY

A. This study presents an RFID-Based Automated Parking System for Revenue Transparency designed to improve payment validation and transaction accuracy at the BzBlock Parking Facility. The system automates the recording of vehicle entry and exit through RFID card scanning conducted by parking staff. Each card is registered upon entry to record time-in and scanned again upon exit to record time-out, allowing the system to automatically compute parking duration and corresponding fees. The system ensures that every transaction is logged and marked with a verified payment status before exit is permitted by staff. By automating fee computation and transaction recording, the system eliminates reuse of previous payments and minimizes revenue discrepancies.

B. **Proponents.** The study enables the proponents to apply their technical knowledge in system development, RFID-based transaction processing, and database management within a real-world environment. It enhances their skills in system analysis, problem-solving, and implementation of automated solutions that address operational and financial challenges in commercial facilities.

Future Proponents. The proposed system serves as a foundation for future research and system enhancement. Future developers may integrate advanced features such as mobile payment platforms, automated receipt generation, predictive revenue analytics, and smart city–compatible parking solutions to further improve operational efficiency and transparency.

Business Owners. The system provides the parking owner with accurate and reliable transaction records, enabling clear visibility of daily, weekly, and monthly parking income. It helps prevent revenue loss caused by unpaid parking, incorrect fee computation, and undocumented transactions, thereby strengthening financial control and supporting informed management decisions.

Parking Facility Management. Parking managers benefit from automated computation of parking fees and centralized transaction logs that reduce manual errors and workload. The system allows management to easily verify payments, review transaction histories, and detect inconsistencies, resulting in improved accountability, efficiency, and overall revenue transparency.

X. DEFINITION OF TERMS

Automated Parking System – *noun* /ˈɔːtəˌmeɪtɪd ˈpɑːrkɪŋ ˈsɪstəm/

A computer-based system that automatically records, computes, and stores parking transactions.

Data Accuracy – *noun* /ˈdeɪtə ˈækjʊrəsi/

The correctness and reliability of recorded parking information within the system.

Fee Computation – *noun* /fiː ˌkɒmpjuˈteɪʃən/

The automated calculation of parking charges based on system-defined rates.

Manual Parking System – *noun* /ˈmænjʊəl ˈpɑːrkɪŋ ˈsɪstəm/

A traditional parking method that relies on paper tickets and manual computation.

Parking Duration – *noun* /ˈpɑːrkɪŋ djʊəˈreɪʃən/

The total length of time a vehicle stays inside the parking facility, calculated from time-in to time-out.

Parking Facility – *noun* /ˈpɑːrkɪŋ fəˈsɪlɪti/

The designated area or building where parking transactions are recorded.

Parking Fee – *noun* /ˈpɑːrkɪŋ fiː/

The amount charged based on the computed parking duration.

Parking Manager – *noun* /ˈpɑːrkɪŋ ˈmænɪdʒər/

Authorized personnel responsible for operating the parking system and validating transactions.

Parking Owner – *noun* /ˈpɑːrkɪŋ ˈoʊnər/

The individual or entity responsible for monitoring parking income and system reports.

Parking Transaction – *noun* /ˈpɑːrkɪŋ trænˈzækʃən/

A recorded event containing vehicle entry time, exit time, parking duration, and fee.

Payment Status – *noun* /ˈpeɪmənt ˈsteɪtəs/

The indicator showing whether a transaction is marked as paid or unpaid.

RFID (Radio Frequency Identification) – *noun* /ˌɑːr.eɪfˈaɪ.diː/

A technology that uses radio waves to transmit data between a reader and a tag.

RFID Card – *noun* /ˈɑːr.eɪfˈaɪ.diː kärd/

A contactless card used to record parking entry and exit information.

Revenue Report – *noun* /ˈrɛvəˌnjuː rɪˈpɔːrt/

A system-generated summary of parking income for a specified period.

Revenue Transparency – *noun* /ˈrɛvəˌnjuː trænsˈpærənsi/

The clear visibility and traceability of all parking transactions and income records.

System Administrator – *noun* /ˈsɪstəm ədˈmɪnɪstreɪtər/

A user with full authority to manage system settings and user access.

System Dashboard – *noun* /ˈsɪstəm ˈdæʃbɔːrd/

A web-based interface displaying transaction data and revenue summaries.

Time-In – *noun* /ˈtaɪm ɪn/

The recorded time when a vehicle enters the parking facility.

Time-Out – *noun* /ˈtaɪm aʊt/

The recorded time when a vehicle exits the parking facility.

Transaction Log – *noun* /trænˈzækʃən lɒg/

A digital record that stores detailed parking transaction information.

CHAPTER 2

This chapter explains how the system was analyzed, designed, and developed. It covers the current system, the proposed system, and the technical components involved in building the new system.

SYSTEM ANALYSIS

A. REQUIREMENTS DOCUMENTATION

The BzBlock Parking Facility continues to experience revenue loss due to weaknesses in its manual parking management process. One of the most common issues occurs when vehicles exit the premises without paying. Since there is no automated payment verification before exit, these unpaid transactions often go unnoticed.

Another recurring issue arises when a customer parks within the allotted time they have already paid for and leaves without exceeding that duration. When the same customer returns on the following day, parking staff may mistakenly allow the vehicle to park for free, assuming that the previous payment is still valid. Due to the absence of a digital transaction history and expiration tracking, staff are unable to verify whether the prior payment was intended only for the previous visit. This misunderstanding contributes to additional revenue loss and inconsistent fee enforcement.

B. CURRENT SYSTEM UTILIZED

The BzBlock Parking Facility currently operates under a fully manual parking management system. Vehicle entry and exit are monitored by parking attendants using paper tickets and cash-based payments. Upon arrival, drivers are issued a handwritten or printed parking ticket indicating the time of entry. Payment is collected manually by staff upon exit, based on estimated parking duration.

There is no computerized system to automatically record time-in and time-out, compute parking fees, or verify whether a vehicle has already paid. As a result, parking attendants rely on personal judgment when determining parking charges and validating payments. This manual process increases the risk of miscalculations, inconsistent fees, and missed collections.

When a customer parks within the paid time limit and exits without exceeding it, the same customer may be allowed to park again on a later date without paying. This occurs because the current system does not maintain a transaction history or time-bound payment validation, making it impossible for staff to confirm whether previous payments are still valid.

THE NEW SYSTEM (SYSTEM DESIGN)

A. REQUIREMENTS ANALYSIS

RESEARCH DESIGN AND METHODOLOGY

The study employed a Research and Development (R&D) design, with the primary objective of designing, developing, and implementing an RFID-Based Automated Parking System for Revenue Transparency at the BzBlock Parking Facility. The proponents adopted the System Development Life Cycle (SDLC) as the development framework, following the phases of analysis, design, development, and testing. This approach ensured a systematic and structured process in creating a functional system that directly addresses the identified problems of inaccurate parking fee computation, and revenue monitoring inefficiencies present in the existing manual parking system.

To evaluate the effectiveness of the proposed system, the proponents utilized a mixed-method research approach combining quantitative and qualitative techniques. The quantitative component involved the use of a five-point Likert Scale questionnaire distributed to drivers who frequently use the parking facility. This instrument gathered measurable data regarding the system's ease of use, efficiency, reliability, and overall user satisfaction.

The qualitative component consisted of system testing, on-site observation, and feedback collection from parking personnel and management. These methods allowed the proponents to assess the system's real-world performance, verify the accuracy of RFID-based time-in and time-out recording, and determine whether the system effectively supports payment validation and prevents unpaid parking through staff-assisted verification.

INSTRUMENT USED IN THE STUDY

To collect data and evaluate the system's effectiveness, the proponents utilized a mixed-method approach consisting of a Likert Scale Questionnaire and interviews. The questionnaire was designed to measure specific aspects of the proposed RFID-based parking system, including the ease of use of the RFID card process, the accuracy of automated parking duration and fee computation, and the effectiveness of the system. A five-point Likert scale was used, where respondents indicated their level of agreement with each statement (1 = strongly disagree, 5 = strongly agree).

The questionnaire was administered to drivers who frequently use the BzBlock Parking Facility, as they are the primary users of the system. The collected responses provided insights into user experience, system usability, satisfaction, and overall acceptance of the RFID-based automated parking solution.

In addition, the proponents conducted interviews with the parking manager using open-ended questions to gather qualitative data regarding current operational challenges, issues related to unpaid parking, and expectations for the proposed system. These interviews provided in-depth insights into parking operations, staff workload, and revenue monitoring concerns. The qualitative findings complemented the quantitative data and supported the evaluation of the system's effectiveness in improving revenue transparency and parking management efficiency.

STATISTICAL TREATMENT OF DATA

The quantitative data gathered from the Likert Scale Questionnaire were analyzed using descriptive statistical methods. Frequency distribution was employed to determine how often each response appeared for every survey item, while percentage analysis was used to identify the proportion of respondents who provided favorable or unfavorable ratings.

To further analyze respondents' perceptions of the system, the weighted mean was computed for each major component of the RFID-based automated parking system, including ease of use, system reliability, accuracy of parking fee computation. The weighted mean was calculated using the following formula:

$$Weighted\ Mean = \frac{\Sigma(f \times x)}{N}$$

Where:

f = frequency of responses

x = numerical value of each response

N = total number of responses

ETHICAL CONSIDERATION

The development of the Automated Card-Based Parking Management System focused on key ethical issues to ensure responsible and fair use of the technology. The proponents emphasized user privacy by collecting only essential information necessary for payment and vehicle entry/exit, and no personal data beyond this is stored. Informed consent was obtained from participants during surveys and interviews, and all feedback was anonymized to protect individual identities.

The system is designed to ensure fair access to parking slots and to prevent any bias in operation, while the automated system controls guarantee transparency and consistency in payment enforcement. Intellectual property rights were respected by accurately citing all external references used in the study. The proponents also considered the possible impact on staff workload, aiming to reduce manual labor without eliminating necessary human assistance for system oversight or handling issues.

To discourage misuse, the system collects only the data required for proper functioning and does not perform surveillance or track user behavior beyond parking sessions. Additionally, energy-efficient components were chosen to minimize environmental impact, and the system was designed for easy accessibility by all users. Throughout the research, the proponents maintained transparency, integrity, and social responsibility, ensuring that the system benefits the parking facility, staff, and drivers ethically and responsibly.

B. INFORMATION MODEL (CONCEPTUAL DESIGN)

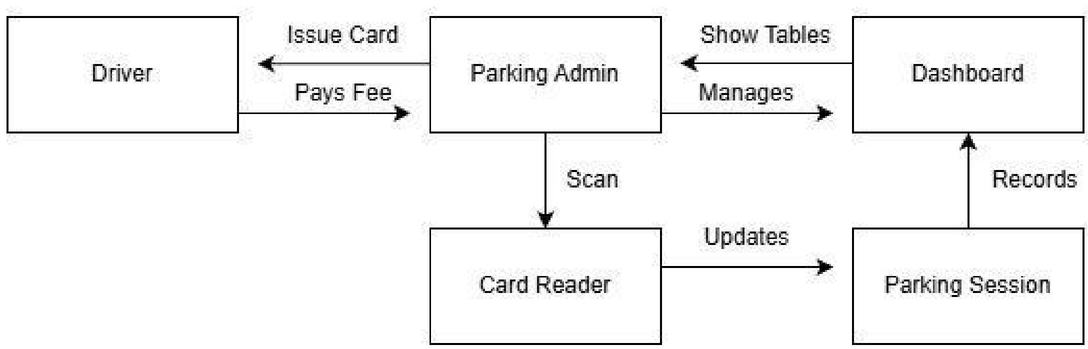


Figure 8: Information Model

The information model represents a parking management system centered on controlled access, payment, and session tracking. A Driver initiates the process by paying a fee to the Parking Admin, who in turn issues a parking card. The Parking Admin manages system operations through a

Dashboard, which provides tabular views of parking data. Card validation is performed via a Card Reader that scans issued cards and updates the corresponding Parking Session. Each parking session records transactional and temporal data, which is subsequently stored and reflected in the Dashboard for monitoring and administrative oversight. The model clearly delineates responsibilities, data flow, and system interactions to support efficient parking operations and accountability.

C. CONTEXT DIAGRAM LEVEL 0 (PHYSICAL DESIGN)

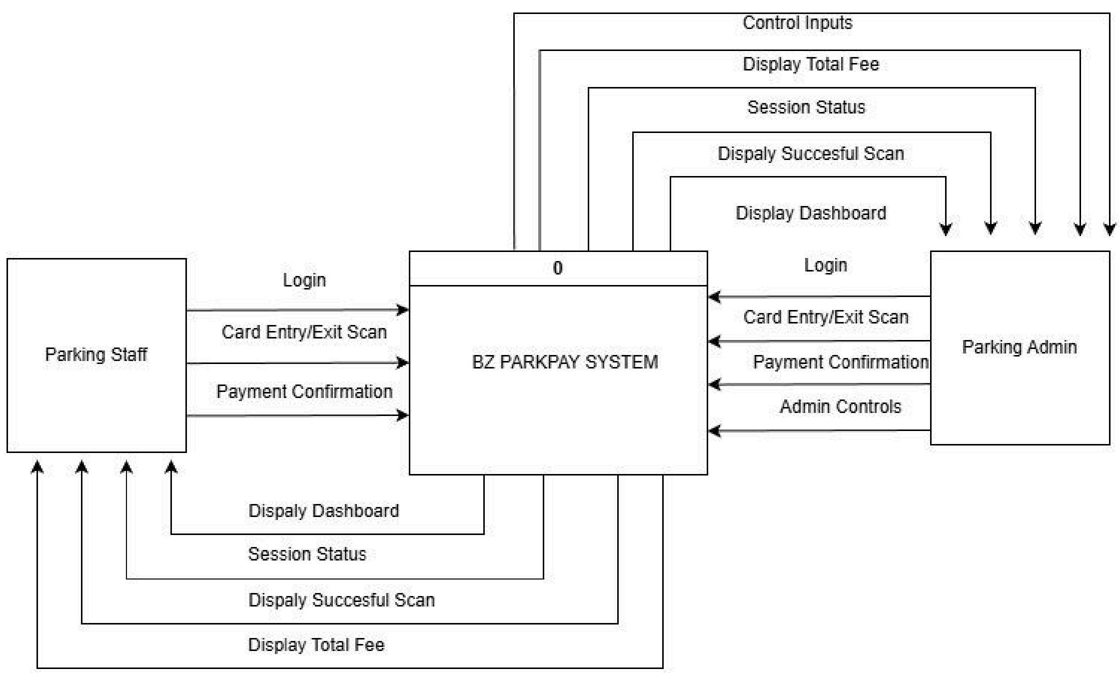


Figure 9: Context Diagram Level 0

The Level 0 Data Flow Diagram (DFD) represents the BZ ParkPay System as a single high-level process and shows its interaction with external entities. The Driver interacts with the system through the Parking Admin by receiving a parking card and paying the parking fee. The Parking Admin and staff operates the system by logging in and performing card entry scans, payment confirmation, and card exit scans, additional Admin Controls only enabled to Administrators. These inputs are processed by the BZ Parking System to manage parking sessions and calculate fees. The system then outputs relevant information to the Parking Admin, including the dashboard display, session status, successful scan confirmation, and the total parking fee.

D. DFD LEVEL 1 (PHYSICAL DESIGN)

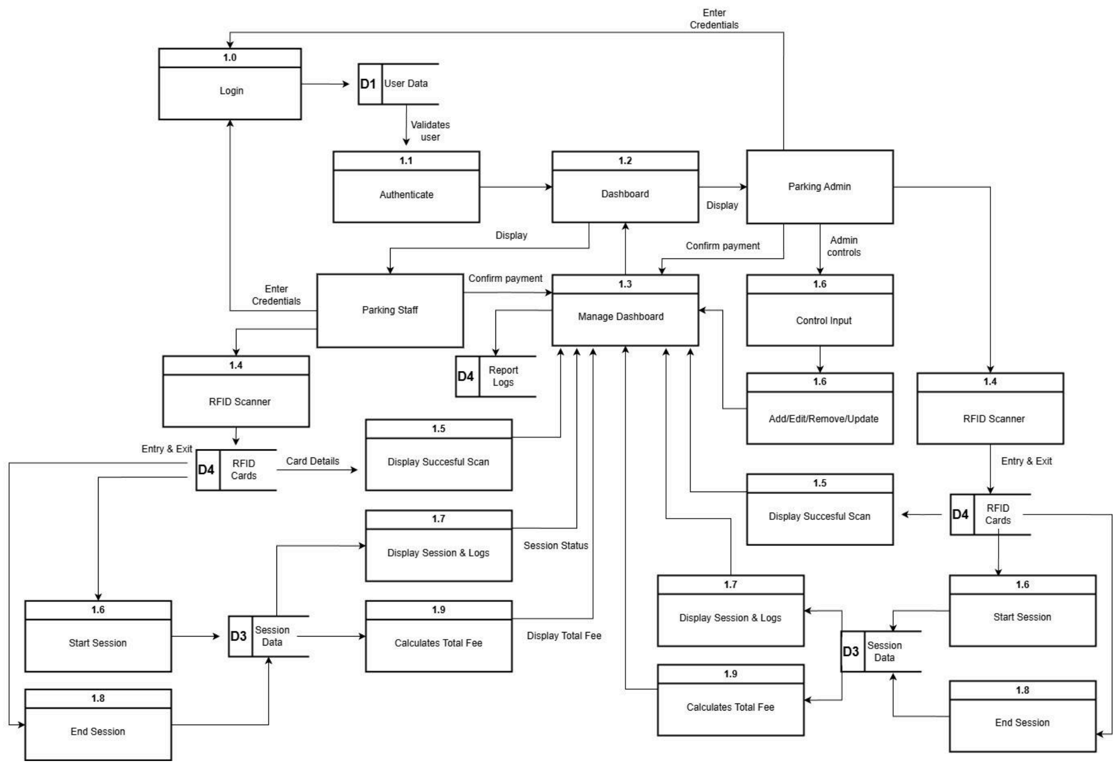


Figure 10: DFD Level 1

The Level 1 Data Flow Diagram (DFD) illustrates the internal processes of the BZ Parking System by detailing how parking activities are handled from entry to exit, along with administrative operations. When a driver arrives, the Parking Admin issues an RFID card, which is scanned at the entrance through the Entrance RFID Reader. The system validates the card against the RFID

Cards data store and, upon successful validation, creates a new parking session that is recorded in the Session Data store. The Parking Admin logs into the system to access the dashboard, where real-time session information is displayed and managed. Through the dashboard, the staff and admin can monitor active and completed sessions, view system logs, manage payments, and perform manual overrides when necessary. The Admin has more control anda re able to configure the system. At the exit point, the driver presents the RFID card, which is scanned by the Exit RFID Reader to retrieve the corresponding parking session. The system then calculates the parking fee and any applicable penalties based on the session duration and updates the payment details in the Payments data store.

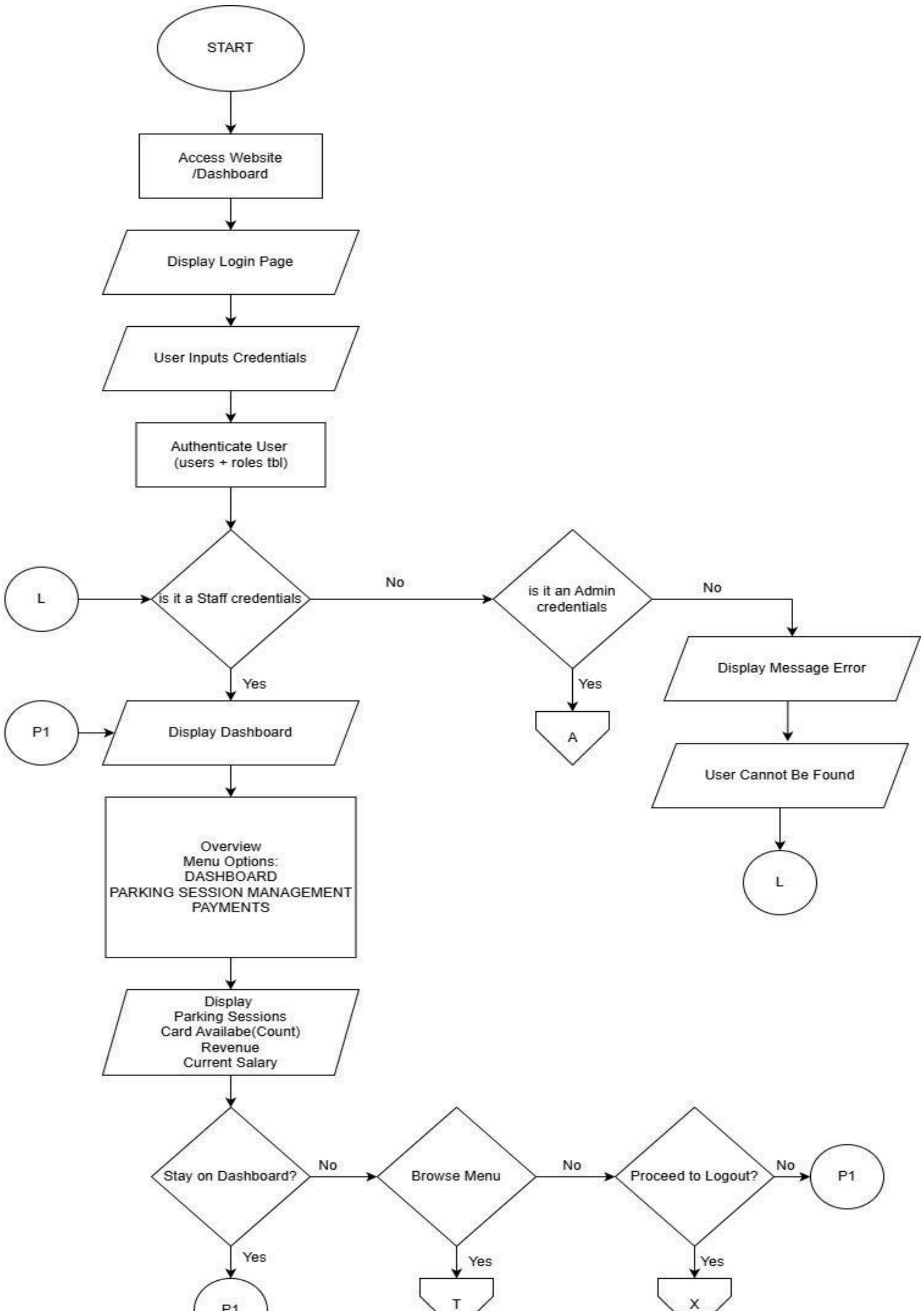
E. ENTITY RELATIONSHIP DIAGRAM/DATABASE CONFIGURATION

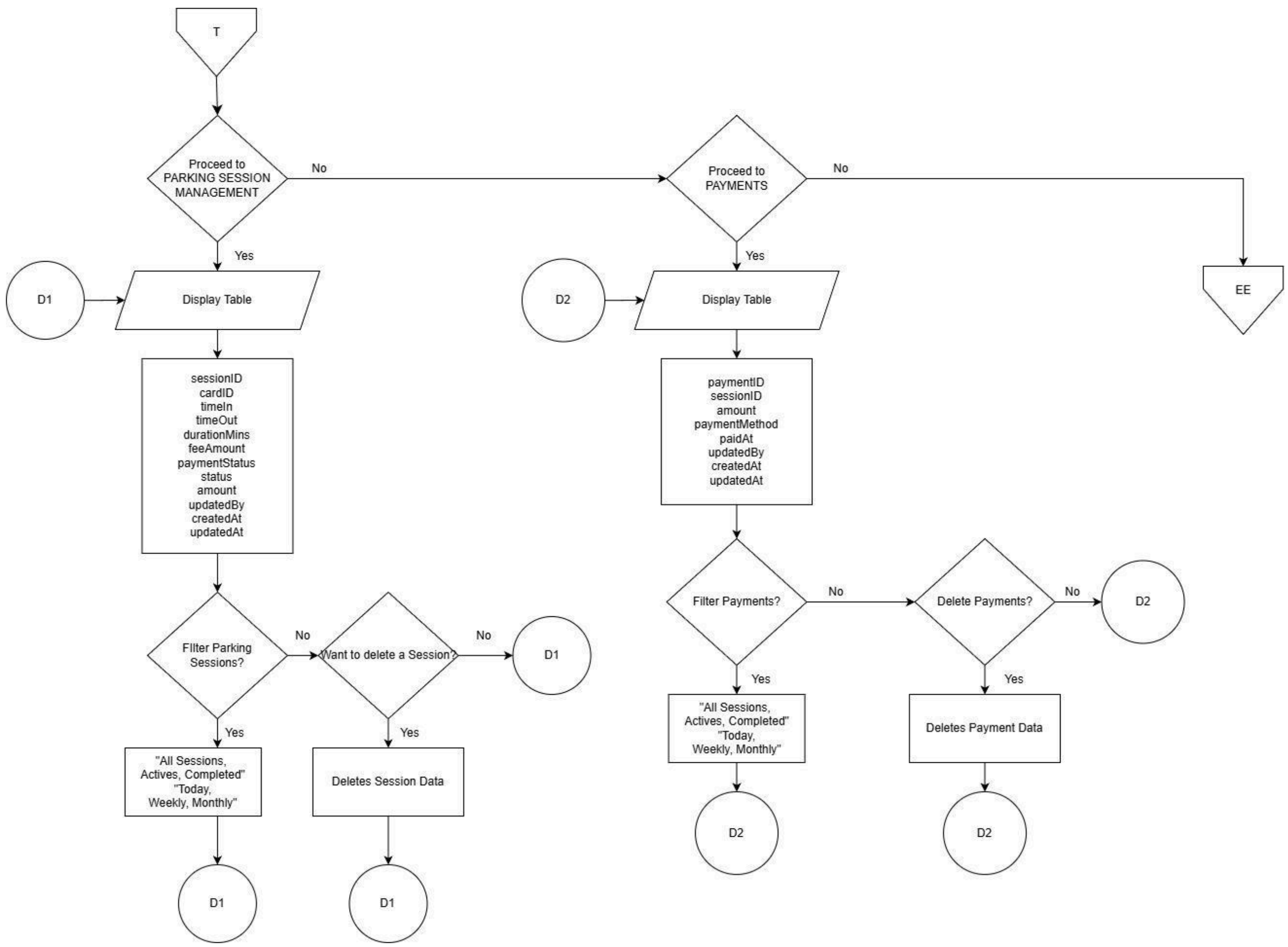
Figure11: Entity Relationship Diagram

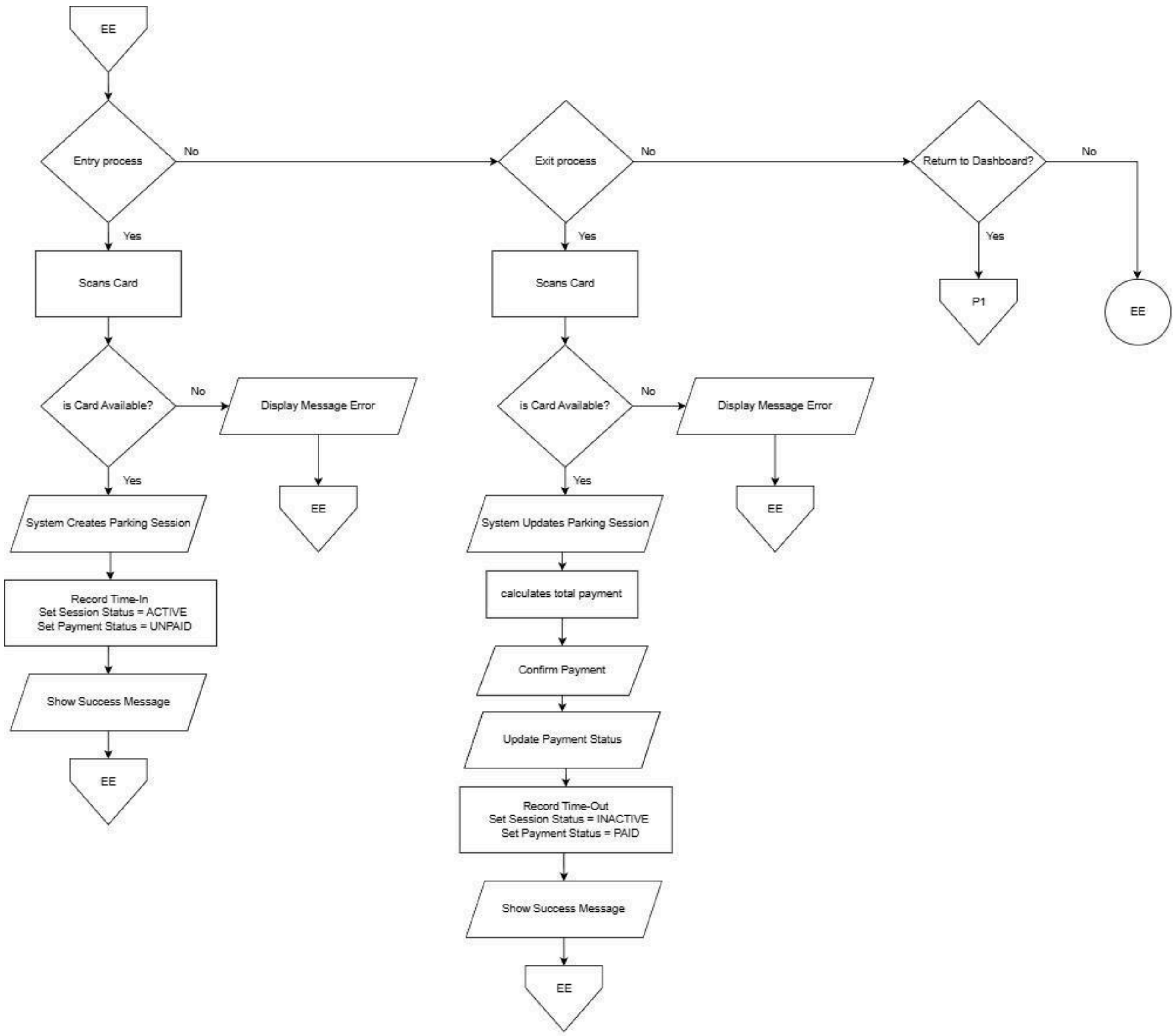
The database design of the BZ Parking System is structured to support access control, parking operations, financial transactions, auditing, and staff management in a normalized and relational manner. The Access Control and

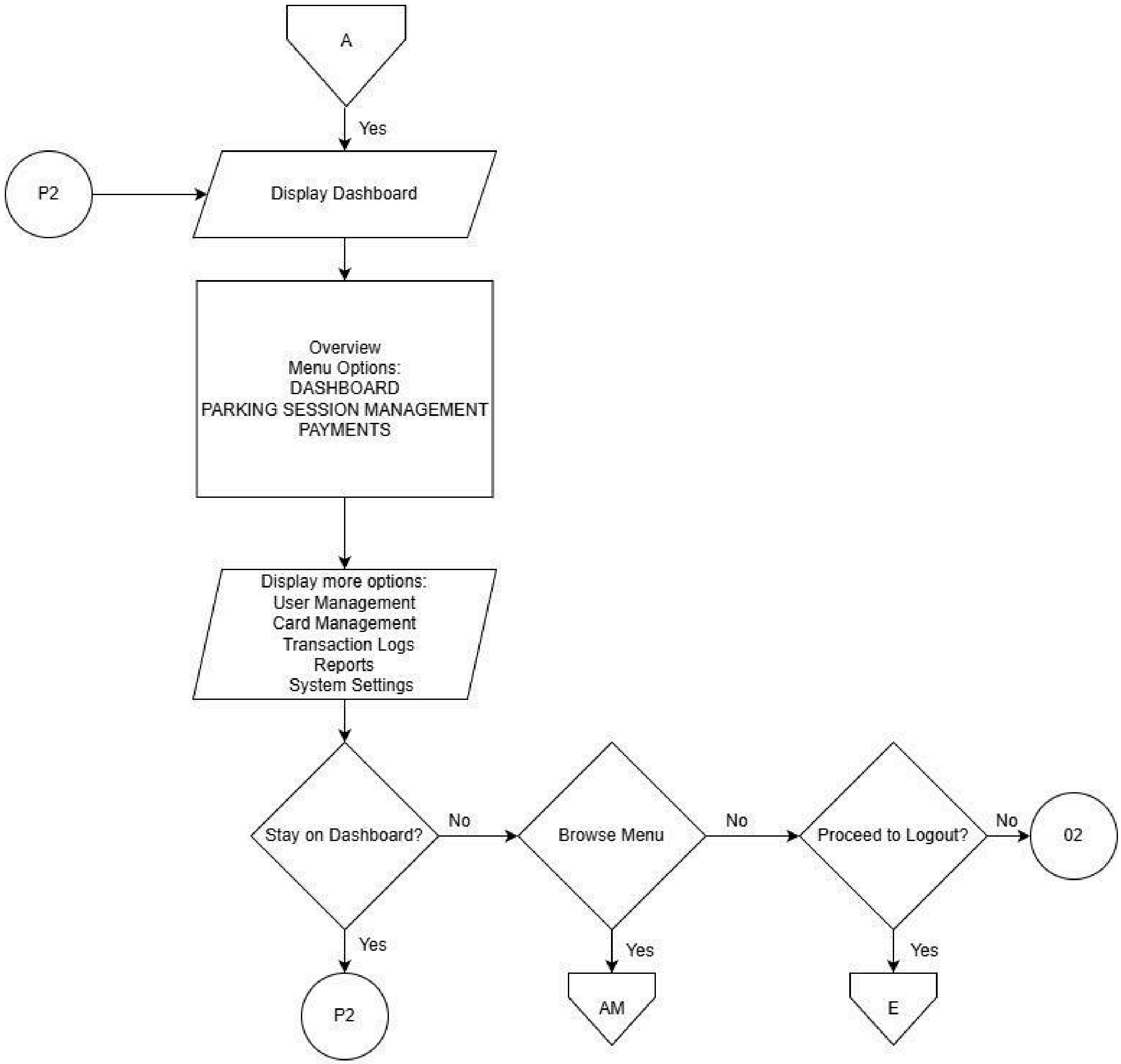
User Management component defines core user authentication and authorization mechanisms. Users are stored in the users table, while role-based access control is implemented through the roles, user_roles, status, and user_status tables, allowing flexible assignment of user roles and account states. Additional user-related information is maintained in the profiles table, while session continuity and security are supported through the refresh_tokens table for token-based authentication. The Parking Core System manages parking operations through RFID-enabled cards and session tracking. The cards table maintains the status and issuance of RFID cards, while parking_sessions records vehicle entry and exit times, duration, calculated fees, payment status, and session state. Each session is linked to the responsible system user to ensure accountability. Payment transactions are stored in the payments table, capturing payment amounts, methods, timestamps, and administrative updates. The Logging and Audit component is handled by the system_logs table, which records system actions, reference entities, and the user responsible for each action, ensuring traceability and operational transparency. Additionally, the Staff Salary Management module supports internal administrative operations through the staff_salary and staff_salary_balance tables, allowing tracking of salary records, payment status, remaining balances, and update history. Overall, the relational structure and defined foreign key relationships ensure data integrity, consistency, and scalability across all functional areas of the system.

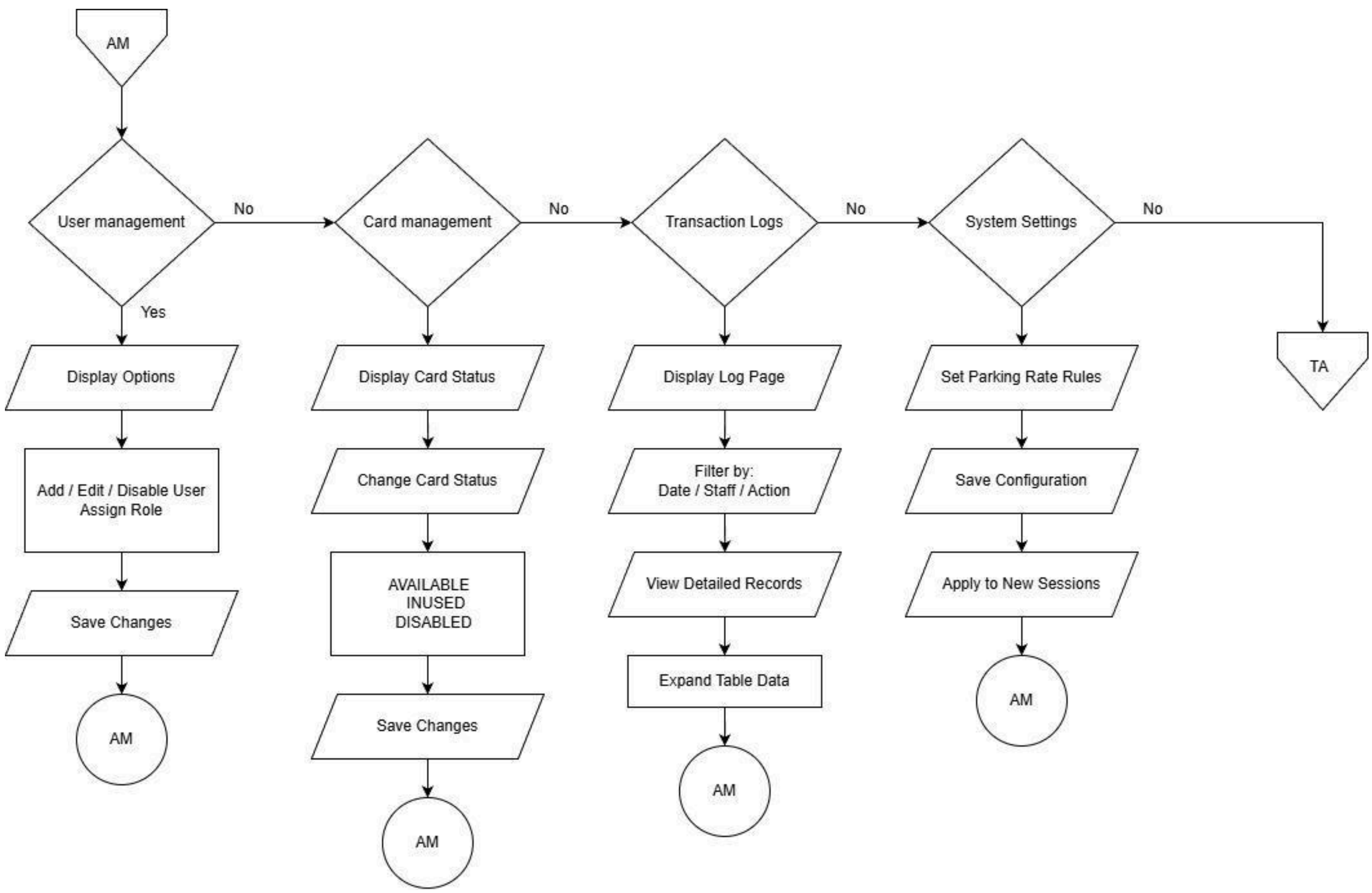
F. FLOWCHART

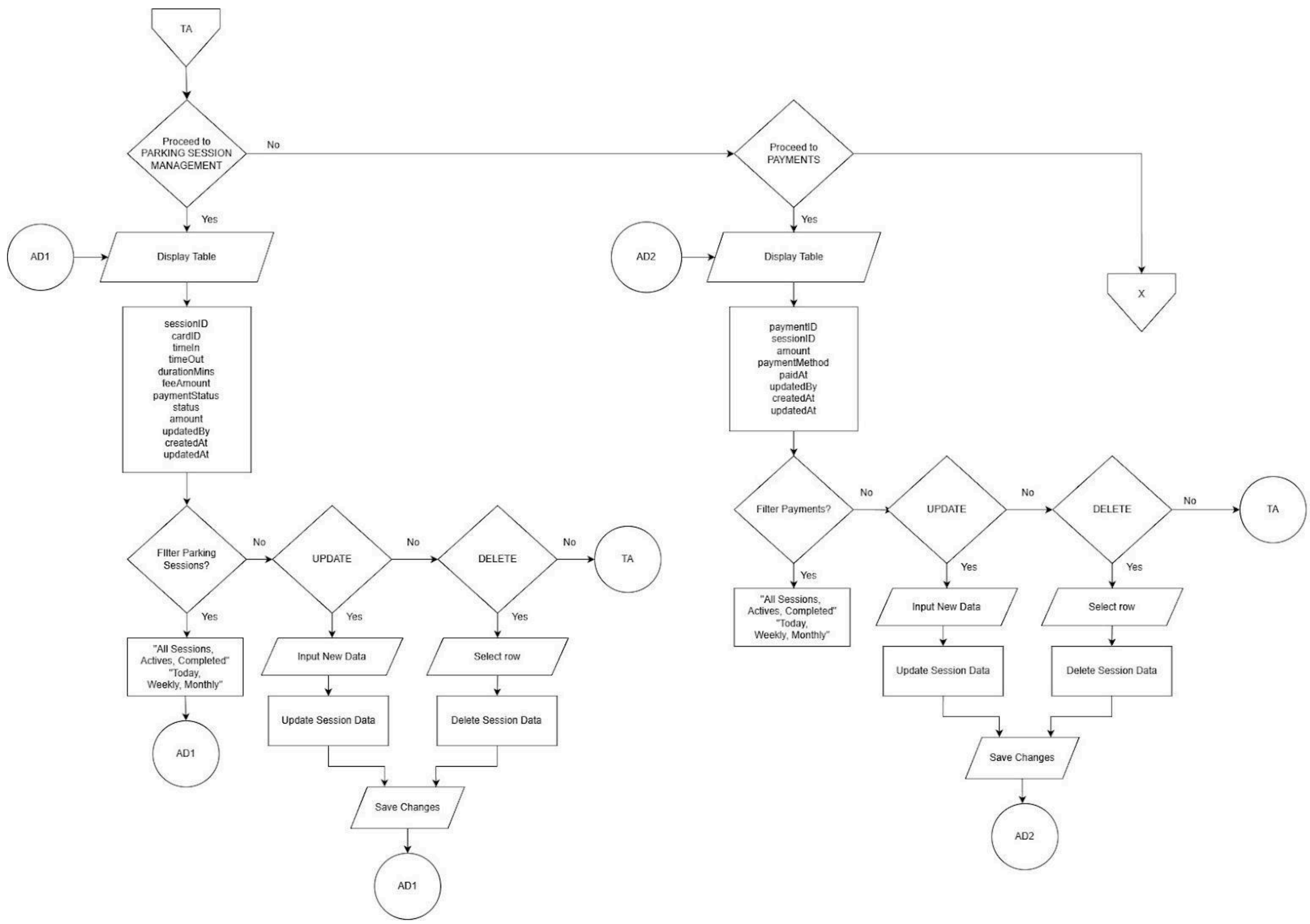



T
Proceed to PARKING SESSION MANAGEMENT
No
Proceed to PAYMENTS
No
Yes
D1
Display Table
D2
Display Table
EE
sessionID
cardID
timeIn
timeOut
durationMins
feeAmount
paymentStatus
status
amount
updatedBy
createdAt
updatedAt
paymentID
sessionID
amount
paymentMethod
paidAt
updatedBy
createdAt
updatedAt
Filter Payments?
No
Delete Payments?
No
D2
Filter Parking Sessions?
No
Want to delete a Session?
No
D1
Yes
"All Sessions, Actives, Completed" "Today, Weekly, Monthly"
Deletes Payment Data
"All Sessions, Actives, Completed" "Today, Weekly, Monthly"
Deletes Session Data
D2
D2
D1
D1


EE
Entry process
No
Yes
Scans Card
is Card Available?
No
Display Message Error
EE
Yes
System Creates Parking Session
Record Time-In
Set Session Status = ACTIVE
Set Payment Status = UNPAID
Show Success Message
EE
Exit process
No
Yes
Scans Card
is Card Available?
No
Display Message Error
EE
Yes
System Updates Parking Session
calculates total payment
Confirm Payment
Update Payment Status
Record Time-Out
Set Session Status = INACTIVE
Set Payment Status = PAID
Show Success Message
EE
Return to Dashboard?
No
Yes
P1
EE


A
Yes
P2
Display Dashboard
Overview
Menu Options:
DASHBOARD
PARKING SESSION MANAGEMENT
PAYMENTS
Display more options:
User Management
Card Management
Transaction Logs
Reports
System Settings
Stay on Dashboard?
No
Browse Menu
No
Proceed to Logout?
No
02
Yes
P2
Yes
AM
Yes
E


AM
User management
No
Yes
Display Options
Add / Edit / Disable User
Assign Role
Save Changes
AM
Card management
No
Display Card Status
Change Card Status
AVAILABLE
INUSED
DISABLED
Save Changes
AM
Transaction Logs
No
Display Log Page
Filter by:
Date / Staff / Action
View Detailed Records
Expand Table Data
AM
System Settings
No
Set Parking Rate Rules
Save Configuration
Apply to New Sessions
AM
TA



TA
Proceed to PARKING SESSION MANAGEMENT
No
Yes
AD1
Display Table
sessionID
cardID
timeIn
timeOut
durationMins
feeAmount
paymentStatus
status
amount
updatedBy
createdAt
updatedAt
Filter Parking Sessions?
No
Yes
"All Sessions, Actives, Completed" "Today, Weekly, Monthly"
AD1
UPDATE
No
Yes
Input New Data
Update Session Data
DELETE
No
Yes
Select row
Delete Session Data
TA
Save Changes
AD1
Proceed to PAYMENTS
Yes
X
AD2
Display Table
paymentID
sessionID
amount
paymentMethod
paidAt
updatedBy
createdAt
updatedAt
Filter Payments?
No
Yes
"All Sessions, Actives, Completed" "Today, Weekly, Monthly"
UPDATE
No
Yes
Input New Data
Update Session Data
DELETE
No
Yes
Select row
Delete Session Data
TA
Save Changes
AD2

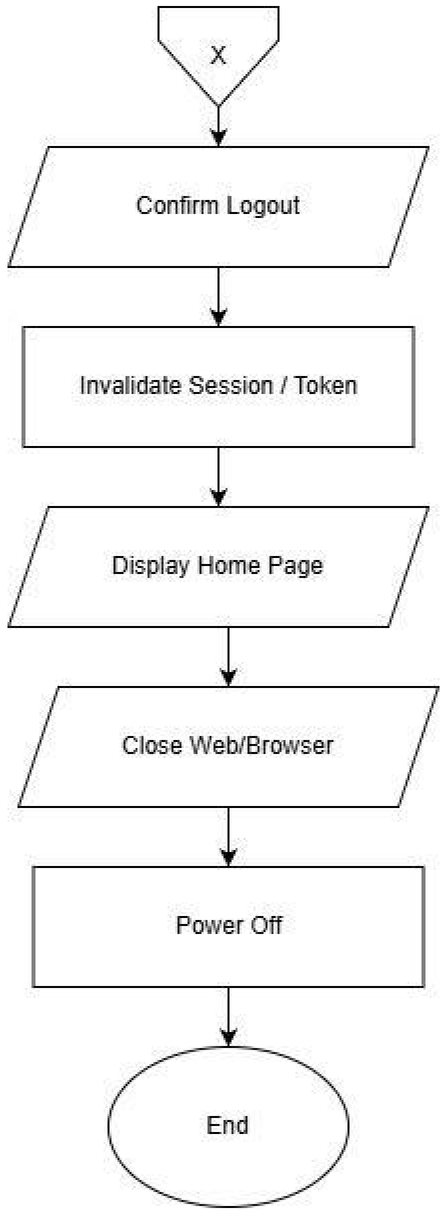


Figure 12: Flowchart

The Parking Management System follows a structured, role-based system flow designed to ensure secure access, efficient parking operations, and controlled administrative management. The system flow begins with user authentication and extends through parking session handling, payment processing, and administrative oversight.Upon accessing the system, users are directed to the login interface where credentials are validated against the system’s user and role records. The authentication process determines whether the user is authorized as a staff member or an administrator. Invalid credentials result in an error message and redirection back to the login page, ensuring system security. Once authenticated, both staff and administrators are directed to a shared dashboard interface. This dashboard provides an overview of current system data, including active parking sessions, available RFID cards, revenue summaries, and other operational indicators. From the dashboard, users may remain on the overview screen, navigate to system modules, or log out of the system.

The Parking Session Management module allows authorized users to view detailed session records, including card identifiers, time-in and time-out values, duration, fee amounts, and payment status. Users may filter session records based on time or status and, where permitted, update or delete session entries. All actions return to the session management interface to maintain workflow continuity.The Payments module enables users to view and manage payment records associated with parking sessions. Payment data may be filtered by date or status, reviewed in tabular form, updated when necessary, or removed if invalid. This module ensures accurate tracking of all financial transactions within the system. The Entry and Exit process is handled through RFID card scanning. During entry, the system verifies card availability, creates a new parking session, records the time-in, and assigns an active session and unpaid payment status. During exit, the system validates the card, calculates the total parking fee based on duration, confirms payment, records the time-out, and updates the session and payment statuses accordingly. Error messages are displayed if invalid or unavailable cards are detected.

Administrators are granted additional system privileges beyond staff access. From the administrative dashboard, administrators may access the User Management module to add, edit, disable users, and assign roles. The Card Management module allows administrators to monitor and update RFID card statuses. The Transaction Logs module provides access to detailed system activity records, supporting filtering and auditing functions. Through System Settings, administrators may configure parking rates and other parameters, which are applied only to newly created parking sessions to preserve transaction consistency. All administrative modules are menu-driven and return to the administrative menu upon completion, ensuring a clear and controlled navigation structure. The system design emphasizes data integrity, role-based access control, operational efficiency, and scalability.

G. ARCHITECTURE DESIGN

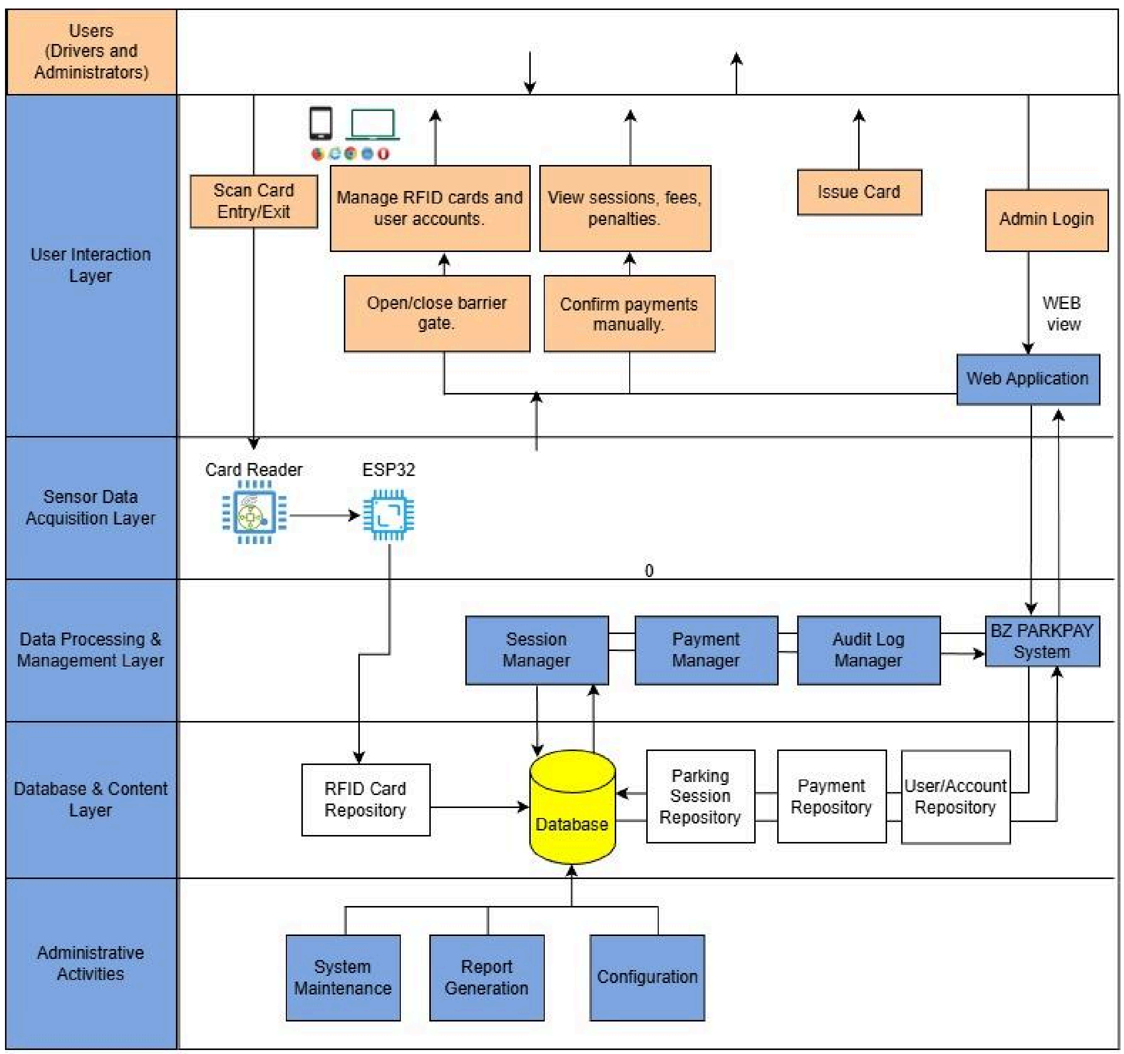


Figure 13: Architecture Design

The architecture design diagram illustrates a layered structure for the BZ Parking System, organized to ensure clear separation of concerns, scalability, and efficient data flow. At the User Interaction Layer, drivers and administrators interact with the system through RFID card scanning at entry and exit points, as well as through a web application interface. This layer supports core user functions such as issuing RFID cards, managing user accounts, viewing parking sessions, fees, and penalties, confirming payments manually, and controlling the barrier gate. The Sensor Data Acquisition Layer handles real-time data collection from physical devices, where the RFID card reader captures card information and transmits it through an ESP32 microcontroller to the backend system. This layer acts as the bridge between hardware components and software services. The Data Processing and Management Layer contains the core business logic of the system. It includes dedicated modules such as the Session Manager, Payment Manager, and Audit Log Manager, which are responsible for managing parking sessions, processing payments, maintaining transaction records, and enforcing system rules. These modules interact with the BZ-Bock System to coordinate system operations and service requests. The Database and Content Layer provides persistent storage through a centralized database supported by specialized repositories, including RFID card, parking session, payment, and user account repositories. This design ensures structured data management and reliable access to operational records.

H. ALGORITHM DESIGN

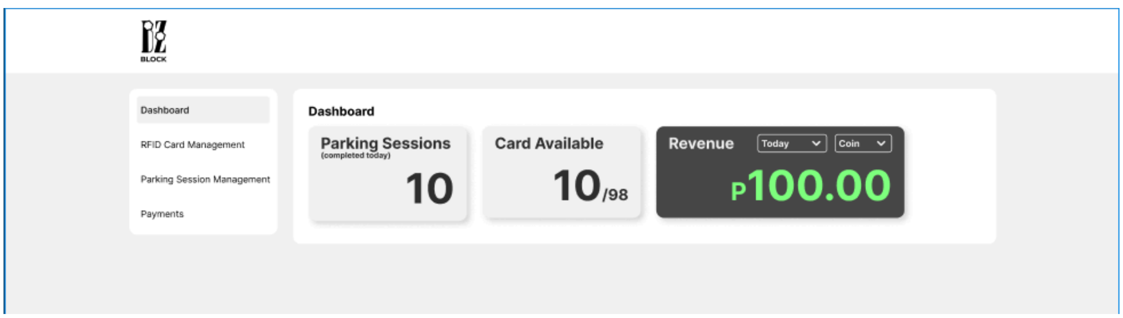


Figure 14: Dashboard

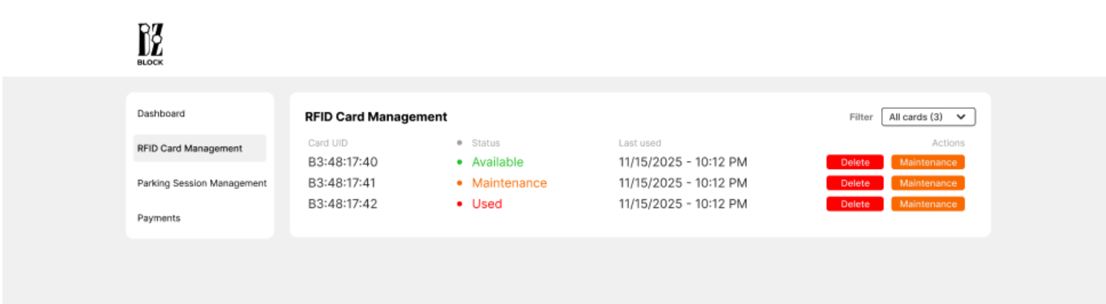


Figure 15: RFID Card Management

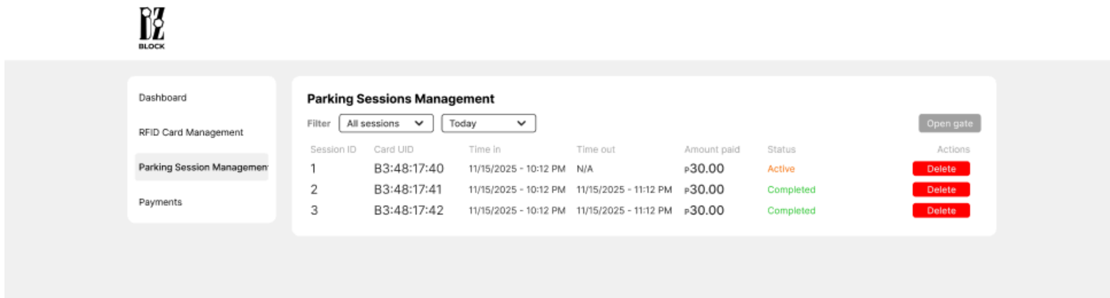


Figure 16: Parking Session Management

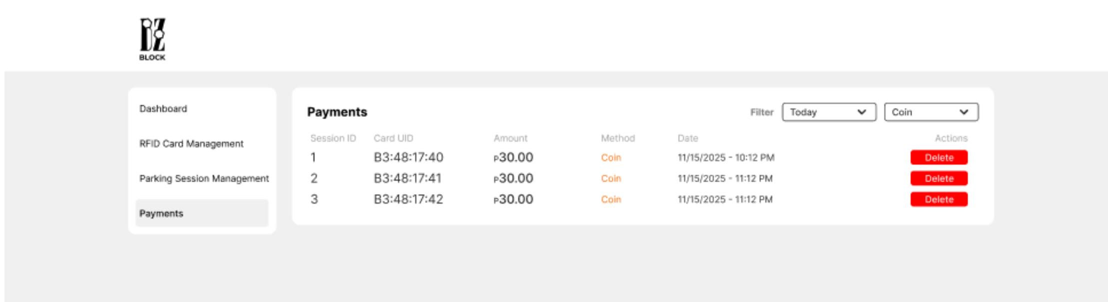


Figure 17: Payments

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